Question 1 [10 pts]: With the indicated link costs, use Dijkstra's shortest-path algorithm to compute the shortest path from x to all network nodes. Show how the algorithm works.

x-z: xz

	y	z	t	u	v	w	N'
1	6	8	∞	∞	3	6	x
2	6	8	7	6		6	x
3							xz

x-t: xvt

	y	z	t	u	v	w	N'
1	6	8	∞	∞	3	6	x
2	6	8	7	6		6	xv
3	6	8	7			6	xv
4							xvt

x-w: xw

	y	z	t	u	v	w	N'
1	6	8	∞	∞	3	6	x
2	6	8	7	6		6	x
3							xw

x-v: xv

	y	z	t	u	v	w	N'
1	6	8	∞	∞	3	6	x
2							xv

x-u: xvu

	y	z	t	u	v	w	N'
1	6	8	∞	∞	3	6	x
2	6	8	7	6		6	xv
3							xvu

x-y: xy

	y	z	t	u	v	w	N'
1	6	8	∞	∞	3	6	x
2	6	8	7	6		6	x
3							xy

Question 2 [5 pts]: Describe how loops in paths can be detected in BGP.

They are detected with the AS_PATH tag, which records the routers an advertisement has visited. When a router receives an advertisement, it checks the AS_PATH and if it sees its own AS path in the list, it detects a loop.

Question 3 [15 pts]: True or false and explain.

a. When an OSPF route sends its link state information, it is sent only to those nodes directly attached to neighbors.

False. In OSPF, LSAs are sent to direct neighbors, which are sent to direct neighbors, et cetera. So technically each wave is only sending to nodes directly attached to neighbors, but the overall route is sent to all nodes.

b. When a BGP router receives an advertised path from its neighbor, it must add its own identity to the received path and then send that new path on to all of its neighbors.

False. It does add its own AS number to the AS_PATH, but it does not send its path back to the neighbor that it got the path from so it doesn't send it to *all* neighbors.

c. A BGP router always chooses the loop-free route with the shortest AS-path length.

False. It chooses the path with the highest Local Preference, and then path length after that. So it's not always picking the shortest AS-path length.

Question 4 [10 pts]: Why do we need both link-level and end-end reliability?

Link-level reliability ensures that data is correctly transmitted across adjacent nodes and can correct some errors, but only works across individual nodes. It cannot account for packets getting lost or not reaching the right destination. End-end reliability ensures that data is correctly delivered from the original sender to the final receiver, but doesn't do link-level checks. It cannot account for the data being corrupted. Having both of those ensures that the data is sent from original sender to receiver, and that it is correct.

Question 5 [10 pts]: Suppose two nodes start to transmit at the same time a packet of length *L* over a broadcast channel of rate *R*. Denote the propagation delay between the two nodes as d_{prop}. Will there be a collision if $d_{prop} < L/R$? Why or why not?

There will be a collision. The time to send a packet is L/R and the time to travel from one node to another is d_{prop} If the time to travel from one node to another is less than the time to send a packet, packets will be received by nodes sooner than they can send them. Basically there's going to be collisions because they're being received faster than they're being sent.

Question 6 [20 pts]: Suppose four active nodes—nodes A, B, C and D—are competing for access to a channel using slotted ALOHA. Assume each node has an infinite number of packets to send. Each node attempts to transmit in each slot with probability p. The first slot is numbered slot 1, the second slot is numbered slot 2, and so on.

a. What is the probability that node A succeeds for the first time in slot 5?

$= P(4 \text{ failures}) \times P(\text{success}) = (1-p)^4 \times p(1-p)^3$

b. What is the probability that some node (either A, B, C or D) succeeds in slot 4?

$= P(\text{any of the 4 succeed in slot 4}) = 4p(1-p)^3$

c. What is the probability that the first success occurs in slot 3?

$= P(3 \text{ failures}) \times P(\text{any of the 4 succeed}) = (1-p(1-p)^3)^2 \times 4p(1-p)^3$

d. What is the efficiency of this four-node system?

efficiency = P(slot has one transmission) = $4p(1-p)^3$

need ¼ success and ¾ failure per node, so $4 \times \frac{1}{4} \times (\frac{3}{4})^3 = {}^{27}/_{64}$

Question 7 [5 pts]: Use your own words to describe the hidden terminal problem in wireless networks.

Say there are two nodes X and Y and a receiver Z. X and Y are out of range of each other, so they can't sense each other's transmissions, but they are both in range of Z. If X and Y transmit to Z at the same time, their signals will collide but X and Y won't be able to detect the collision and resulting data corruption. Basically, this is a problem about not being able to sense other transmissions and causing data collisions as a result.

Question 8 [10 pts]: Why is an ARP query sent within a broadcast frame? Why is an ARP response sent within a frame with a specific destination MAC address?

An ARP query is sent within a broadcast frame because the sender needs to send it to a specific address but doesn't know which receiver has the address. Broadcasting it means all receivers get the query.

An ARP response is only sent to a specific destination because the sender only needs to let the querier know that they have the address. Broadcasting that would be unnecessary traffic since the responder knows which address asked.

Question 9 [15 pts]: Consider three LANs interconnected by two routers.

a. Assign IP addresses to all the interfaces except for interfaces of switches. For Subnet 1 use addresses of the form 192.168.1.xxx; for Subnet 2 uses addresses of the form 192.168.2.xxx; and for Subnet 3 use addresses of the form 192.168.3.xxx.

Subnet 1 (192.168.1.0/24)

Host A: 192.168.1.2
Host B: 192.168.1.3
Rtr 1 to Sub 1: 192.168.1.1

Subnet 2 (192.168.2.0/24)

Host C: 192.168.2.2
Host D: 192.168.2.3
Rtr 1 to Sub 2: 192.168.2.1
Rtr 2 to Sub 2: 192.168.2.4

Subnet 3 (192.168.3.0/24)

Host E: 192.168.3.2
Host F: 192.168.3.3
Rtr 2 to Sub 3: 192.168.3.1

b. Assign MAC addresses to all the adapters (interfaces).

Subnet 1

Host A: 00:1A:2B:3C:4D:5E
Host B: 00:1A:2B:3C:4D:6F
Rtr 1 to Sub 1: 00:1A:2B:3C:4D:1A

Subnet 2

Host C: 00:2A:3B:4C:5D:6E
Host D: 00:2A:3B:4C:5D:7F
Rtr 1 to Sub 2: 00:2A:3B:4C:5D:1B
Rtr 2 to Sub 2: 00:2A:3B:4C:5D:2C

Subnet 3

Host E: 00:3A:4B:5C:6D:7E
Host F: 00:3A:4B:5C:6D:8F
Rtr 2 to Sub 3: 00:3A:4B:5C:6D:3D

c. Consider sending an IP datagram from Host E to Host D. Suppose all the ARP tables are up to date. Enumerate all the steps.

1. Host E checks if Host D is on the same subnet, but 192.168.3.0 ≠ 192.168.2.0 so it determines Host D is on a different subnet.
2. Host E checks the routing table and determines that packet should be sent to router 2.
3. Host E looks up the MAC address for router 2 (192.168.3.1) in its ARP table and finds router 2's MAC address (00:3A:4B:5C:6D:3D).
4. Host E creates the package with source IP 192.168.3.2 and destination IP 192.168.2.3; and source MAC: 00:3A:4B:5C:6D:7E and destination MAC: 00:3A:4B:5C:6D:3D.
5. Host E sends package to the switch in Subnet 3, which checks the destination MAC address and forwards it to Router 2.
6. Router 2 receives the package, checks the destination IP address, and decides the package should go to Subnet 2.
7. Router 2 checks if it has the MAC address for Host D in its ARP table and finds the address.
8. Router 2 updates the package with source MAC 00:2A:3B:4C:5D:2C and destination MAC 00:2A:3B:4C:5D:7F (same source and destination IPs as step 4).
9. Router 2 sends the package to the switch in Subnet 3, which checks the destination MAC address and sends it to Host D.
10. Host D receives the package, verifies that the destination IP address matches its IP address, and processes the package.